

Boolean algebras without free families

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Introduction

A family $\{\alpha_i : i \in I\}$ of elements of a Boolean algebra B , is said to be *free* if for every choice of finite by many distinct elements $i_1, i_2, \dots, i_n$ of I , it follows that $\bigwedge_{\sigma=1}^n \varepsilon_\sigma \alpha_{i_\sigma} \neq 0$ where $\varepsilon_\sigma \alpha_{i_\sigma}$ denotes either α_{i_σ} or else $1 - \alpha_{i_\sigma}$.

Balcar and Franěk in [3] have proved that every complete Boolean algebra B contains a free family $\{\alpha_i : i \in I\}$ with cardinality equal to $|B|$ (cardinality of B). Tsarpalias and the author in [1] and [2] have proved some results on the existence of free families, in a class of Boolean algebras B , with cardinality equal to $|B|$. Similar results, are also given by Shelah in [5].

Our purpose, in this paper, is to prove some results in the opposite direction to the ones described above. We deal here with the existence of a Boolean algebra B with the property: every free family of elements of B has cardinality less than $|B|$.

More precisely in the first part of the paper we prove:

THEOREM A. *Let λ be an uncountable regular cardinal and B a Boolean algebra generated by a family $\mathcal{D} = \{\alpha_\xi : \xi < \lambda\}$ such that for every filter $\mathcal{F}$ of elements of B , it follows that $|\mathcal{F} \cap \mathcal{D}| < \lambda$. Then there is no free family $\{d_\xi : \xi < \lambda\}$ in B .*

In the second part, using theorem A, we prove

THEOREM B (G.C.H.). *For every singular cardinal α with $\beta = cf(\alpha) < \alpha$ there is a Boolean algebra B_α such that $|B_\alpha| = \alpha^+$, B_α has the β^+ chain condition (i.e. every family of pairwise disjoint non-zero elements of B_α has cardinality less than β^+) and there is no free family $\{d_\xi : \xi < \alpha^+\}$ in B_α .*

The last theorem gives an affirmative answer to a conjecture of Shelah [5].

§0.

DEFINITIONS AND PRELIMINARY RESULTS. Let X be a set; a family $\{(A_i, B_i) : i \in I\}$ of pairs of subsets of X is *independent* if $A_i \cap B_i = \emptyset$ and for every $n < \omega$, $\{\varepsilon_1, \dots, \varepsilon_n\} \in \{-1, 1\}^n$, $i_1, \dots, i_n$ pairwise different elements of I ,

$$\bigcap_{\kappa=1}^n \varepsilon_{\kappa} A_{i_{\kappa}} \neq \emptyset,$$

where we have set $(1) \times A_{i_{\kappa}} = A_{i_{\kappa}}$ and $(-1) \times A_{i_{\kappa}} = B_{i_{\kappa}}$.

In a Boolean algebra B , a family $\{\alpha_i : i \in I\} \subseteq B$ is *free* if the corresponding (via Stone's duality theorem) family $\{(V_{\alpha_i}, V_{\alpha_i}^c) : i \in I\}$ of clopen subsets of $S(B)$ is independent.

It is more convenient for us to use topological space language instead of the corresponding terminology of Boolean algebras. However it is easy to translate statements and proofs that appear here into Boolean algebras terminology, via Stone's duality theorem.

For a topological space X we denote by $S(X)$ the least cardinal κ such that every family $\{U_i : i \in I\}$ of non-empty open subsets of X has cardinality strictly less than κ . We also let $d(X)$ be the least cardinal α such that some dense subset of X has cardinality equal to α .

A topological space X has α -*caliber* (α -*precaliber*) if every family $\{U_{\xi} : \xi < \alpha\}$ of non-empty open subsets of X has a subfamily $\{u_{\xi} : \xi \in I\}$ such that $|I| = \alpha$ and $\bigcap_I U_{\xi} \neq \emptyset$ ($\{U_{\xi} : \xi \in I\}$ has the finite intersection property).

Let X be a completely regular topological space. We denote by βX the Čech-Stone compactification of X ; and for a compact space X we denote by $G(X)$ the Gleason space of X , namely the unique extremally disconnected compact space that is mapped continuously onto X by an irreducible map. From the properties of $G(X)$ follows that $G(X)$ has α -caliber iff X has α -caliber.

Let $\{X_i : i \in I\}$ be a family of topological spaces and κ be a cardinal. We denote by

$$(\Pi_I X_i)_{\alpha}$$

the topological space that has a basis of open sets consisting of sets of the form $\Pi_I V_i$ where V_i is open in X_i and the set $\{i \in I : V_i \neq X_i\}$ has cardinality less than α .

More results about the Gleason space and products with strong topologies can be found in [4]. In the same textbook, the proof of the following Erdős-Rado result is given.

0.1. LEMMA (Δ -systems lemma). Let α, κ be cardinals with α regular, $\alpha > \kappa$,

and if $\beta < \alpha$, then $\lambda < \kappa$ implies $\beta^\lambda < \alpha$. Then every family $\{A_\xi : \xi < \alpha\}$ of sets with $|A_\xi| < \kappa$ for all $\xi < \alpha$ contains a subfamily $\{A_\xi : \xi \in I\}$ such that $|I| = \alpha$ and $A_{\xi_1} \cap A_{\xi_2} = A$ for $\xi_1, \xi_2 \in I$, $\xi_1 \neq \xi_2$.

§1.

1.1. DEFINITION. Let S be a set and $\{(U_\xi, V_\xi) : \xi < \alpha\}$ be an independent family of subsets of S .

For every natural number $n < \omega$ we define a family $\{(U_\xi^n, V_\xi^n) : \xi < \alpha\}$ inductively by the next rule.

For $n = 1$ we set $U_\xi^n = U_\xi$, $V_\xi^n = V_\xi$ for all $\xi < \alpha$. If for $n = \kappa$ the family $\{(U_\xi^\kappa, V_\xi^\kappa) : \xi < \alpha\}$ has been defined, then we choose a family $\{(\xi_\sigma, j_\sigma) : \sigma < \alpha\}$ such that for $\sigma_1 < \sigma_2$ we have $\xi_{\sigma_1} < j_{\sigma_1} < \xi_{\sigma_2} < j_{\sigma_2}$, and we set

$$\begin{aligned} U_{\sigma}^{\kappa+1} &= U_{\xi_\sigma}^\kappa \cap V_{j_\sigma}^\kappa \\ V_{\sigma}^{\kappa+1} &= U_{j_\sigma}^\kappa \cap V_{\xi_\sigma}^\kappa. \end{aligned}$$

The inductive definition is complete.

A trivial remark is that for every $n < \omega$ the family $\{(U_\xi^n, V_\xi^n) : \xi < \alpha\}$ is independent.

1.2. THEOREM. Let S be a compact totally disconnected space and α a regular uncountable cardinal. Assume that S has a subbasis $\{(A_\xi, A_\xi^c) : \xi < \alpha\}$ of non-empty open and closed sets with $A_\xi \neq S$ for all $\xi < \alpha$ and such that for every $I \subset \alpha$ with $|I| = \alpha$ $\bigcap_{i \in I} A_\xi = \emptyset$. Then there is no independent family $\{(U_\xi, V_\xi) : \xi < \alpha\}$ with U_ξ, V_ξ non-empty open-and-closed sets and $U_\xi \cap V_\xi = \emptyset$ for all $\xi < \alpha$.

Proof. Let us assume the contrary; then there is an independent family $\{(U_\xi, V_\xi) : \xi < \alpha\}$. Our purpose is to prove that there is a natural number $n_0 < \omega$ such that for the family $\{(U_\xi^{n_0}, V_\xi^{n_0}) : \xi < \alpha\}$ either $\bigcap_{\xi < \alpha} U_\xi^{n_0} = \emptyset$ or else, $\bigcap_{\xi < \alpha} V_\xi^{n_0} = \emptyset$, which obviously contradicts to our assumption.

Step 1. The structure of U_ξ, V_ξ for $\xi < \alpha$.

By passing, to a subfamily of $\{(U_\xi, V_\xi) : \xi < \alpha\}$ with the same cardinality, we can assume that the original family satisfies the following conditions:

(i) there are natural numbers $\mu < \nu$ such that for every $\xi < \alpha$,

$$U_\xi = \bigcup_{i=1}^{\mu} \Lambda_i^\xi, \quad V_\xi = \bigcup_{i=\mu+1}^{\nu} \Lambda_i^\xi$$

where Λ_i^ξ is a basic open-and-closed set generated from subbasis $\{A_\xi, A_\xi^c : \xi < \alpha\}$,

(ii) for each $1 \leq i \leq \nu$ there is an integer $\lambda_i > 0$, $(\varepsilon_1, \dots, \varepsilon_{\lambda_i}) \in \{-1, 1\}^{\lambda_i}$, and a basic open-and-closed set K_i in S , such that for every $\xi < \alpha$

$$\Lambda_i^\xi = K_i \cap \left(\bigcap_{\kappa=1}^{\lambda_i} \varepsilon_\kappa A_{\xi(\kappa,i)} \right),$$

(iii) if $\xi_1 \neq \xi_2$, $i_1, i_2 \in \{1, \dots, \nu\}$, $\kappa_1 \in \{1, \dots, \lambda_{i_1}\}$, $\kappa_2 \in \{1, \dots, \lambda_{i_2}\}$,

then

$$A_{\xi_1(\kappa_1, i_1)} \neq A_{\xi_2(\kappa_2, i_2)}.$$

(If the original family does not satisfy one of the three properties, then using the regularity of α and the Erdős-Rado Δ -systems lemma, we can find a subfamily with the desired properties that has cardinality equal to α .)

Step 2: The structure of U_ξ^n, V_ξ^n for $n < \omega$ and $\xi < \alpha$.

Assume that the independent family $\{(U_\xi, V_\xi) : \xi < \alpha\}$ satisfies properties (i), (ii) and (iii) as above; let $n < \omega$ and let $\{(U_\sigma^n, V_\sigma^n) : \sigma < \alpha\}$ be a family that is obtained from the $\{(U_\xi, V_\xi) : \xi < \alpha\}$ as in Definition 1.1. Then each pair U_σ^n, V_σ^n is a finite union of some intersections of the sets

$$\{\Lambda_\xi^i : \xi < \alpha : i = 1, \dots, \nu\}.$$

More precisely, for each $\sigma < \alpha$ and $\varepsilon = 1$ or $\varepsilon = -1$ we have

$$\varepsilon U_\sigma^n = \bigcup_{j=1}^{\kappa_\sigma} M_j^\sigma$$

and

$$M_j^\sigma = \left(\bigcap_{i \in \Gamma_{(j,\sigma)}} K_i \right) \cap \left(\bigcap_{\xi \in B_{(j,\sigma)}} \varepsilon_\xi^i A_\xi \right), \quad (*)$$

where $\Gamma_{(j,\sigma)} \subset \{1, \dots, \nu\}$ and $B_{(j,\sigma)} \subset \alpha$ are such that

$$B_{(j_1, \sigma_1)} \cap B_{(j_2, \sigma_2)} = \emptyset$$

for $\sigma_1 < \sigma_2$, $j_1 \leq \kappa_{\sigma_1}$ and $j_2 \leq \kappa_{\sigma_2}$.

1.3. DEFINITION. Let $n < \omega$ and $\{(U_\sigma^n, V_\sigma^n) : \sigma < \alpha\}$ be a family as above. Let $\sigma < \alpha$ and $M_i^\sigma \subset \varepsilon U_\sigma^n$ be a basic clopen set with the form (*) as above. We define the *characteristic set* of M_i^σ to be the set $\Gamma_{(i, \sigma)} \subset \{1, \dots, \nu\}$ such that

$$M_i^\sigma = \left(\bigcap_{i \in \Gamma_{(i, \sigma)}} K_i \right) \cap \left(\bigcap_{\xi \in B_{(i, \sigma)}} \varepsilon_\xi^i A_\xi \right).$$

The *characteristic number* of the family $\{(U_\sigma^n, V_\sigma^n) : \sigma < \alpha\}$ is the smaller integer d such that there is a set $I \subset \alpha$ with $|I| = \alpha$, $\Gamma_1 \subset \{1, \dots, \alpha\}$, and for every $\sigma \in I$, $M_i^\sigma \subset \varepsilon_\sigma U_\sigma^n$ exists such that

- (1) Γ_I is the *characteristic set* of M_i^σ for all $\sigma \in I$
- (2) $\bigcap_{\sigma \in I} M_i^\sigma \neq \emptyset$.

1.4. LEMMA. Let S be a set and $\{U_\xi : \xi < \alpha\}$ a family of subsets of S such that each U_ξ is a union $\bigcup_{i \in I_\xi} V_i$ and $\bigcap_\alpha U_\xi \neq \emptyset$. Then there is a family $\{i_\xi : \xi < \alpha\}$ such that $i_\xi \in I_\xi$ and $\bigcap_\alpha V_{i_\xi} \neq \emptyset$.

Proof. Let $s_0 \in \bigcap_\alpha U_\xi$. For each $\xi < \alpha$ we choose $i_\xi \in I_\xi$ such that $s_0 \in V_{i_\xi}$. Obviously, the family $\{i_\xi : \xi < \alpha\}$ has the desired property.

1.5. LEMMA. Let $\{(U_\sigma^n, V_\sigma^n) : \sigma < \alpha\}$ be the family in step 2 of the proof of Theorem 1.2. Also, assume that for some fixed $\varepsilon \in \{-1, 1\}$ and for all $\sigma < \alpha$ there exists $M_i^\sigma \subset \varepsilon U_\sigma^n$ and $\Gamma \subset \{1, \dots, \nu\}$ such that

- (i) $\bigcap_\alpha M_i^\sigma \neq \emptyset$
- (ii) Γ is the *characteristic set* of M_i^σ for all $\sigma < \alpha$. Then the set $\mathcal{D} = \{\sigma < \alpha : \exists M_i^\sigma \subset (-\varepsilon)U_\sigma^n \text{ with } \Gamma \text{ the characteristic set of } M_i^\sigma\}$,

has cardinality less than α .

Proof. First we remark that if a family $\{M_i^\sigma : \sigma < \alpha\}$ satisfies properties (i) and (ii) then every set M_i^σ , for which Γ is the characteristic set, has the form

$$M_i^\sigma = \left(\bigcap_{i \in \Gamma} K_i \right) \cap \left(\bigcap_{\xi \in B_{(i, \sigma)}} (-1) A_\xi \right).$$

In fact, if $M_{i_1}^{\sigma_1}, M_{i_2}^{\sigma_2}$ have the same characteristic set, then they are obtained as the intersection of the same type of original basic clopen sets A_ξ^i , and if there is a $\xi_{\sigma_1} \in B_{(i_1, \sigma_1)}$ with $\varepsilon_{\xi_{\sigma_1}} = 1$, then there is a $\xi_{\sigma_2} \in B_{(i_2, \sigma_2)}$ with $\varepsilon_{\xi_{\sigma_2}} = 1$. But, this implies that for every $\sigma < \alpha$ there is A_{ξ_σ} with $M_i^\sigma \subset A_{\xi_\sigma}$; and if $\sigma_1 < \sigma_2$ then $\xi_{\sigma_1} \neq \xi_{\sigma_2}$. From this fact it follows that $\bigcap_\alpha A_{\xi_\sigma} \neq \emptyset$, a contradiction.

Assume, now that the cardinality of the set $\mathcal{D}$ is equal to α . By the previous remark,

$$M_j^\sigma \subset \left(\bigcap_{i \in \Gamma} K_i \right) \cap \left(\bigcap_{\xi \in B(i, \sigma)} A_\xi \right),$$

for every $\sigma \in \mathcal{D}$.

Since $\bigcap_{\sigma \in \mathcal{D}} M_j^\sigma \neq \emptyset$, it follows from lemma 1.4 that there is a set $I \subset \alpha$ with $|I| = \alpha$ and $\bigcap_{\xi \in I} A_\xi \neq \emptyset$, a contradiction. The proof is complete.

1.6. LEMMA. *Let d_n be the characteristic number of a family $\{(U_\sigma^n, V_\sigma^n) : \sigma < \alpha\}$. Then a family $\{(U_\xi^{n+1}, V_\xi^{n+1}) : \xi < \alpha\}$ can be obtained from the previous one that has characteristic number greater than or equal to d_{n+1} .*

Proof. With no loss of generality we can assume that

(a) if $\Gamma \subset \{1, \dots, \nu\}$ and there exist $\sigma_0 < \alpha$ and $\varepsilon \in \{-1, 1\}$ such that Γ is the characteristic set of some $M_{j_0}^{\sigma_0} \in \varepsilon U_{\sigma_0}^n$, then for each $\sigma < \alpha$ there is $M_j^\sigma \subset \varepsilon U_\sigma^n$ with characteristic set the set Γ ;

(b) if Γ and ε are given and for $\sigma < \alpha$ there exists $M_j^\sigma \subset \varepsilon U_\sigma^n$ with characteristic set Γ , then either there is a family $\{M_{j_\sigma}^\sigma : M_{j_\sigma}^\sigma \subset U_\sigma^n\}$ such that Γ is the characteristic set of $M_{j_\sigma}^\sigma$ and $\bigcap_\alpha M_{j_\sigma}^\sigma \neq \emptyset$, or, $\bigcap_{\sigma \in I} M_{j_\sigma}^\sigma = \emptyset$ for all $I \subset \alpha$ with $|I| = \alpha$ and for every choice of $\{M_{j_\sigma}^\sigma : M_{j_\sigma}^\sigma \subset \varepsilon U_\sigma^n\}$.

Let $\{(\sigma_\xi, \tau_\xi) : \xi < \alpha\}$ be a choice of indices with $\sigma_{\xi_1} < \tau_{\xi_1} < \sigma_{\xi_2} < \tau_{\xi_2}$ for all $\xi_1 < \xi_2 < \alpha$. Setting, now $U_\xi^{n+1} = U_{\sigma_\xi}^n \cap V_{\tau_\xi}^n$ and $V_\xi^{n+1} = V_{\sigma_\xi}^n \cap U_{\tau_\xi}^n$ we remark that if d_{n+1} is the characteristic number of the family $\{(U_\xi^{n+1}, V_\xi^{n+1}) : \xi < \alpha\}$ then $d_{n+1} \geq d_n$.

Assume that $d_{n+1} = d_n$. Then there is a set $\mathcal{D} \subset \alpha$ with $|\mathcal{D}| = \alpha$ and $\varepsilon \in \{-1, 1\}$ such that $\bigcap_{\xi \in \mathcal{D}} G_\xi^j \neq \emptyset$ for a suitable choice of $G_\xi^j \subset \varepsilon U_\xi^{n+1}$.

Each $G_\xi^j = M_{j_1}^{\sigma_\xi} \cap M_{j_2}^{\tau_\xi}$, so that $\bigcap_{\xi \in \mathcal{D}} M_{j_1}^{\sigma_\xi} \neq \emptyset$ and $\bigcap_{\xi \in \mathcal{D}} M_{j_2}^{\tau_\xi} \neq \emptyset$; and since the characteristic number of $\{(U_\sigma^n, V_\sigma^n) : \sigma < \alpha\}$ is d_n , it follows that there is a set $\mathcal{D}' \subset \mathcal{D}$ with $|\mathcal{D}'| = \alpha$ and subsets Γ_1, Γ_2 of $\{1, \dots, \nu\}$ such that for each $\xi \in \mathcal{D}$, Γ_1 is the characteristic set of $M_{j_1}^{\sigma_\xi}$ and Γ_2 is the characteristic set of $M_{j_2}^{\tau_\xi}$. Also, $|\Gamma_1| = |\Gamma_2|$. Since the characteristic number of $\{(U_\sigma^n, V_\sigma^n) : \sigma < \alpha\}$ is d_n it follows that $\Gamma_1 = \Gamma_2 = \Gamma$. But from (b) it follows that there is a family $\{M_{j_1}^\sigma : \sigma < \alpha\}$ such that $\bigcap_\alpha M_{j_1}^\sigma \neq \emptyset$, $M_{j_1}^\sigma \subset U_\sigma^n$ and the characteristic set of $M_{j_1}^\sigma$ is set Γ . However, by (a) and lemma 1.5 there is no $M_i^\sigma \subset V_\sigma^n$ whose characteristic is the set Γ , a contradiction.

Proof of theorem completed. From last lemma we conclude that for some $n \leq \nu + 1$ the characteristic number of the family $\{(U_\sigma^n, V_\sigma^n) : \sigma < \alpha\}$ will be equal

to ν . Then either $\bigcap_{\alpha} U_{\sigma}^n \neq \emptyset$ or $\bigcap_{\alpha} V_{\sigma}^n = \emptyset$ (otherwise we contradict lemma 1.5). So the family $\{(U_{\xi}, V_{\xi}) : \xi < \alpha\}$ is not independent. The proof is complete.

1.7. Remark. Under some additional set-theoretical assumptions the proof of the above theorem is almost trivial. More precisely if we assume that $\alpha^{<\alpha} < 2^{\alpha}$ then we can prove that with the assumptions of theorem, $|S| \leq \alpha^{<\alpha}$. (In fact, for every $s \in S$ we set $U_s = \{\xi : s \in A_{\xi}\}$, and for $s_1 \neq s_2$ it follows that $U_{s_1} \neq U_{s_2}$ and $|U_s| < \alpha$.) On the other hand if there is an independent family $\{(U_{\xi}, V_{\xi}) : \xi < \alpha\}$ of closed sets in a compact space S then $|S| \geq 2^{\alpha}$, and the proof is complete.

1.8. COROLLARY. *Let X be a compact space with Souslin number $S(X) = \kappa$ (i.e. every family of pairwise disjoint open non-empty subsets of X has cardinality less than κ), and assume that X does not have α -caliber for some regular uncountable cardinal α . Then there is a compact totally disconnected space Y with $S(Y) = \kappa$, $w(Y) = \alpha$ and without an independent family of clopen sets with cardinality α .*

Proof. If we denote by GX the Gleason's space of X , then $S(GX) = \kappa$ and GX does not have α -caliber. Applying, theorem 1.2 to GX , we get a space Y with the desired properties as a quotient space of GX .

1.9. Remark. Balcar and Franěk have proved in [3] that every compact extremally disconnected space S has an independent family with cardinality equal to topological weight of S . Thus, we cannot find an extremally disconnected space that satisfies the conclusion of corollary 1.8.

QUESTION: Assume that α is a regular uncountable cardinal and let X be a compact totally disconnected space with $w(X) \geq \alpha$ and α -caliber. Does X have an independent family of cardinality equal to α ?

§2.

This section is devoted to the construction of examples of topological spaces which, in conjunction with theorem 1.2 and G.C.H., allow us to answer affirmatively a question of Shelah [5].

2.1. LEMMA. *Let X be a set with topologies $\mathcal{T}_1, \mathcal{T}_2$ such that there is a base for $\mathcal{T}_1$ consisting of $\mathcal{T}_2$ -compact set, and $|X| = d(X, \mathcal{T}_1) = \alpha$. Then $(X, \mathcal{T}_1)$ does not have precaliber α .*

Proof. Let $X = \{x_\xi : \xi < \alpha\}$ be a well-ordering of the elements of X . For every $\xi < \alpha$ there is $V_\xi \in \mathcal{T}_1$, such that V_ξ is $\mathcal{T}_2$ -compact and $\{x_\xi : \xi \leq \xi\} \cap V_\xi = \emptyset$. If $(X, \mathcal{T}_1)$ has precaliber α , then there is $I \subset \alpha$, $|I| = \alpha$, such that $\{V_\xi : \xi \in I\}$ has the finite intersection property. Since V_ξ is $\mathcal{T}_2$ -compact for $\xi \in I$, it follows that there exists $x \in V_\xi$ for $\xi \in I$. Then $x = x_{\xi_0}$ for some $\xi_0 < \alpha$. If $\xi_1 \in I$ and $\xi_1 > \xi_0$, then

$$x \notin V_{\xi_1};$$

a contradiction.

2.2. LEMMA. (G.C.H.). *Let β be an infinite cardinal. Then, the space $X = (\{0, 1\})^\beta$ has $cf(\beta)^+$ caliber.*

Proof. Let $\{U_\xi : \xi < cf(\beta)^+\}$ be a family of non-empty open subsets of X . With no loss of generality we assume that for all $\xi < cf(\beta)^+$, $U_\xi = \{\varepsilon_{I_\xi}\} \times \{0, 1\}^{\beta \setminus I_\xi}$ where $I_\xi \subset \beta$ with $|I_\xi| < cf(\beta)$ and $\varepsilon_{I_\xi} \in \{0, 1\}^{I_\xi}$.

We apply Erdős-Rado Δ -systems lemma to the family $\{I_\xi : \xi < cf(\beta)^+\}$ and find $\mathcal{D} \subset cf(\beta)^+$, $|\mathcal{D}| = cf(\beta)^+$, and $I \subset \beta$ such that if $\xi_1 \neq \xi_2 \in \mathcal{D}$ then $I_{\xi_1} \cap I_{\xi_2} = I$. Choosing, $\mathcal{D}_1 \subset \mathcal{D}$ with $|\mathcal{D}_1| = |\mathcal{D}|$ such that for $\xi_1, \xi_2 \in \mathcal{D}_1$, $\pi_I(\varepsilon_{I_{\xi_1}}) = \pi_I(\varepsilon_{I_{\xi_2}})$, we easily verify that $\bigcap_{\xi \in \mathcal{D}_1} U_\xi \neq \emptyset$.

2.3. THEOREM (G.C.H.). *Let κ be a regular uncountable cardinal and β a (singular) cardinal, such that $cf(\beta) < \kappa < \beta$. Then, there is a compact Hausdorff extremally disconnected space Ω such that $S(\Omega) = \kappa$ and Ω does not have caliber β^+ .*

Proof. By G.C.H. and the fact that β is a singular cardinal, there is a family $\{\beta_\sigma : \sigma < cf(\beta)\}$ of cardinals such that

$$\sum_{\sigma < cf(\beta)} \beta_\sigma = \beta$$

β_σ is infinite regular cardinal for $\sigma < cf(\beta)$,

$\beta_{\sigma^{++}} \leq \beta_{\sigma+1}$ for $\sigma < cf(\beta)$, and

$\prod_{\sigma < \lambda} \beta_\sigma \leq \beta_\lambda$ for λ limit ordinal such that $0 < \lambda < cf(\beta)$.

Let $\{B_\sigma : \sigma < cf(\beta)\}$ be a family of subsets of β such that

$$|B_\sigma| = \beta_\sigma \text{ for } \sigma < cf(\beta),$$

and

$$B_\sigma \cap B_\tau = \emptyset \text{ for } \sigma < \tau < cf(\beta).$$

We define a tree $(T = \bigcup_{\sigma < cf(\beta)} T_\sigma, \alpha)$ consisting of elements of $\beta^{[2]}$ in the following way.

We set $T_0 = \beta_0^{[2]}$. For $\sigma < cf(\beta)$ we let $\{B_{\sigma+1,i} : i < \beta_\sigma\}$ be such that

$$|B_{\sigma+1,i}| = \beta_{\sigma+1} \text{ for } i < \beta_\sigma,$$

and

$$B_{\sigma+1,i} \cap B_{\sigma+1,j} = \emptyset \text{ for } i < j < \beta_\sigma.$$

We set

$$T_{\sigma+1} = \bigcup_{i < \beta_\sigma} (B_{\sigma+1,i}^{[2]}).$$

Assume that we have defined T_σ inductively such that

$$T_\sigma \subset B_\sigma^{[2]},$$

$$|T_\sigma| = \beta_\sigma.$$

Let $T_\sigma = \{s_i^{(\sigma)} : i < \beta_\sigma\}$ be a one-to-one well-ordering of T_σ . Furthermore, if $s \in T_\sigma$ and $t \in T_{\sigma+1}$ then

$$s \alpha t \text{ if and only if there is } i < \beta_\sigma \text{ such that } s = s_i^{(\sigma)} \text{ and } t \in B_{\sigma+1,i}^{[2]}.$$

Let λ be a limit ordinal, $0 < \lambda < cf(\beta)$ and assume that $\{T_\sigma, \sigma < \lambda\}$ has been defined, together with the ordering α , so that

$$\left(\bigcup_{\sigma < \lambda} T_\sigma, \alpha \right) \text{ is a tree,}$$

and

$$T_\sigma = \{s : \{t : t \alpha s\} \text{ has order type } \sigma\}.$$

Let $\mathcal{G}_\lambda$ be the set of all branches of length λ of this tree, and let $|\mathcal{G}_\lambda| = \beta_\lambda^*$. It is clear that

$$\beta_\lambda^* \leq \prod_{\sigma < \lambda} \beta_\sigma \leq \beta_\lambda.$$

Let $\mathcal{G}_\lambda = \{\sum_i : i < \beta_\lambda^*\}$ be a one-to-one well-ordering of $\mathcal{G}_\lambda$. Let $\{B_{\lambda,i} : i < \beta_\lambda^*\}$ be

such that

$$\begin{aligned} B_\lambda &= \bigcup_{i < \beta_\lambda^*} B_{\lambda,i}, \\ |B_{\lambda,i}| &\geq 2 \text{ for } i < \beta_\lambda^*, \text{ and} \\ B_{\lambda,i} \cap B_{\lambda,j} &= \emptyset \text{ for } i < j < \beta_\lambda^*. \end{aligned}$$

We set

$$T_\lambda = \bigcup_{i < \beta_\lambda^*} (B_{\lambda,i}^{[2]}),$$

and we define the ordering α as follows: if $s \in T_\sigma$ for some $\sigma < \lambda$, and $t \in T_\lambda$ then $s \alpha t$ if and only if there is $i < \beta_\lambda^*$ such that

$$s \in \Sigma_i, \text{ and } t \in b_{\lambda,i}^{[2]}.$$

This completes the definition of the tree

$$\left(T = \bigcup_{\sigma < cf(\beta)} T_\sigma, \alpha \right).$$

Let $\mathcal{G}$ be the set of all branches of T . We set

$$\Sigma(\sigma) = \Sigma \cap T_\sigma,$$

and

$$H_{1,\sigma} = \beta \setminus H_{2,\sigma}$$

$$H_{2,\sigma} = \bigcup_{\sigma < \tau < cf(\alpha)} \bigcup_{s \in T_\tau} s$$

for $\sigma < cf(\beta)$.

For $s \in T$, let K_s be the "anti-diagonal" of $\{0, 1\}^s$ i.e., if $s\{\zeta, \eta\}$ then

$$K_s = \{((\zeta, 1), (\eta, 0)), ((\zeta, 0), (\eta, 1))\}.$$

We next define the topological space X . As a set X is $\{0, 1\}^\beta$. The topology $\mathcal{T}$ on X is defined by a subbasis that consists of

(a) all subsets of $\{0, 1\}^\beta$ that are open-and-closed in the $cf(\beta)$ -box topology $(\{0, 1\}^\beta)_{cf(\beta)}$, and

(b) the family $\{V_\Sigma : \Sigma \in \mathcal{G}\}$, where

$$V_\Sigma = \prod_{s \in \Sigma} K_s \times \{0, 1\}^{\beta \setminus \bigcup_{s \in \Sigma} s}.$$

We note that $(X, \mathcal{T})$ has (a sub-base and hence) a base consisting of sets that are closed (and hence compact) in the usual (product) topology of $\{0, 1\}^\beta$. Thus X is a completely regular Hausdorff space.

Every basic open-and-closed set V of X has the form:

$$V = U \cap \bigcap_{i=1}^k V_{\Sigma_i},$$

where U is a (basic) open-and-closed set of $(\{0, 1\}^\beta)_{cf(\beta)}$ and $\Sigma_1, \dots, \Sigma_k \in \mathcal{G}$; we say in this case that V is determined by $(U; \Sigma_1, \dots, \Sigma_k)$. We say that V is separated at level σ_0 (for some $\sigma_0 < cf(\beta)$) if

- (a) $\sum_{j_1}(\sigma_0) \cap \sum_{j_2}(\sigma_0) = \emptyset$ for $1 \leq j_1 < j_2 \leq \kappa$, and
- (b) if $F \subset cf(\beta)$, $|F| < cf(\beta)$ is the set on which V depends, then $F \cap H_{2, \sigma_0} = \emptyset$.

If V is separated at level σ_0 , then there is an open-and-closed set W (in the $cf(\beta)$ -box topology of $(\{0, 1\}^\beta)_{cf(\beta)}$), with $W \subset \{0, 1\}^{H_{1, \sigma_0}}$ such that

$$V = W \times \bigcap_{j=1}^k \left(\prod_{\sigma_0 < \sigma < cf(\beta)} K_{\Sigma_j(\sigma)} \times \{0, 1\}^{H_{2, \sigma_0} \setminus \bigcup_{\sigma_0 < \sigma < cf(\beta)} \Sigma_j(\sigma)} \right),$$

CLAIM 1. $S(X) = cf(\beta)^+$ (in fact X has property $K_{cf(\beta)^+, 2}$).

Indeed, let $\{V_\eta : \eta < cf(\beta)^+\}$ be a family of basic sets in X . Let V_η be determined by $(U_\eta; \sum_1^{(\eta)}, \dots, \sum_k^{(\eta)})$ for $\eta < cf(\beta)^+$. We assume without loss of generality that there is $\sigma_0 < cf(\beta)$, such that V_η is separated at σ_0 for all $\eta < cf(\beta)^+$. Set $\pi_{H_{1, \sigma_0}}(V_\eta) = W_\eta$ and note that W_η is open-and-closed in $cf(\beta)$ -box topology for all $\eta < cf(\beta)^+$; since $(\{0, 1\}^\beta)_{cf(\beta)}$ has calibre $cf(\beta)^+$ (lemma 2.2), it follows that there is $I \subset cf(\beta)^+$, with $I = cf(\beta)^+$, such that $\{W_\eta : \eta \in I\}$ has the finite intersection property.

It now follows easily that $\{V_\eta : \eta \in I\}$ has the 2-intersection property. In fact, if $\eta_1, \eta_2 \in I$ then, since every branch of T that determines V_{η_2} at a level greater than or equal to σ_0 , it follows that $V_{\eta_1} \cap V_{\eta_2} \neq \emptyset$.

CLAIM 2. $d(X) = |X|$.

Suppose that $d(X) < |X|$, and hence by G.C.H., $d(X) \leq \beta$; there is a set

$\{x_\xi : \xi < \beta\} \subset X$ that is dense in X . Choose $\{s_\sigma : \sigma < cf(\beta)\}$ inductively such that

$$\begin{aligned} s_\sigma &\in T_\sigma \quad \text{for } \sigma < cf(\beta), \\ s_\sigma &< s_\tau \quad \text{for } 0 < \tau < cf(\beta), \end{aligned}$$

and for every $\xi < \beta$ there is $\sigma < cf(\beta)$ such that $x_\xi \notin K_{s_{\sigma+1}} \times \{0, 1\}^{\beta \setminus s_{\sigma+1}}$.

Indeed, let $\sigma < cf(\beta)$, and suppose that s_τ have been defined for $\tau < \sigma$. We will define $s_{\sigma+1}$.

For $\zeta \in B_{\sigma+1,i}$ we set $A_\zeta = \{\xi \in B_\sigma : x_\xi(\zeta) = 1\}$. Then $A_\zeta \in \mathcal{P}(B_\sigma)$; and since

$$|\mathcal{P}(B_\sigma)| = 2^{\beta_\sigma} = \beta_\sigma^+ < |B_{\sigma+1,i}|,$$

it follows that there are $\zeta_1, \zeta_2 \in B_{\sigma+1,i}$, $\zeta_1 \neq \zeta_2$, such that

$$x_\xi(\zeta_1) = x_\xi(\zeta_2) \quad \text{for } \xi \in B_\sigma.$$

We set

$$s_{\sigma+1} = \{\zeta_1, \zeta_2\}.$$

If λ is a limit ordinal, $\lambda < cf(\beta)$, we choose any s_λ with $s_\lambda > s_\sigma$ for all $\sigma < \lambda$. We finally set

$$\Sigma = \{s_\sigma : \sigma < cf(\beta)\} \in \mathcal{G}.$$

Then it is clear that

$$V_\Sigma \cap \{x_\xi : \xi < \beta\} = \emptyset,$$

in contradiction to the assumed density of $\{x_\xi : \xi < \beta\}$.

It now follows from the last claim and lemma 2.1, that X does not have precaliber β^+ .

Construction of the space Ω . We set $Y = \beta X$, the Cech-Stone compactification of X . Let Ω be the Gleason space of Y . Then, we easily check that $S(\Omega) = S(Y) = S(X) = cf(\beta)^+ \leq \kappa$, and Ω does not have β^+ -caliber.

Next corollary follows immediately from the previous theorem and corollary 1.8 and answers Shelah's conjecture affirmatively.

2.4. COROLLARY (G.C.H.). *Let β, κ cardinals with β singular and $cf(\beta) < \kappa$.*

Then there is a compact totally disconnected space Ω such that $S(\Omega) \leq \kappa$, $w(\Omega) = \beta^+$ and every independent family $\{(A_i, B_i) : i \in I\}$ of clopen subsets of Ω has cardinality less than β^+ .

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